

1. Stern-Gerlach



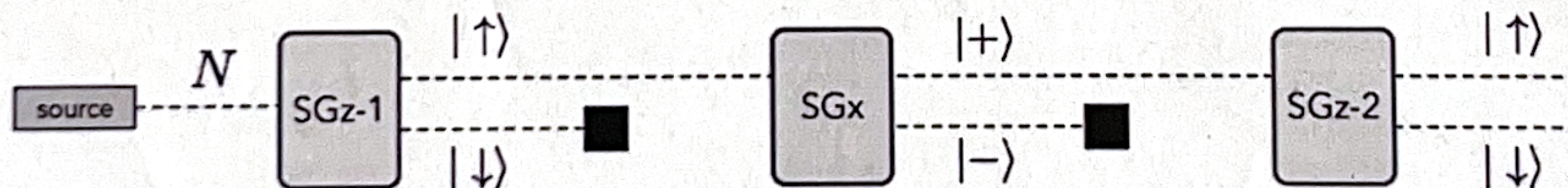
The left side of the picture above gives a schematic representation of the Stern-Gerlach experiment. The source sends an electron in the quantum state

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \quad \text{or} \quad |\psi_2\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$$

through a magnetic field along the z -axis (SGz) to give rise to two possible outcomes, one with the electron in the $|\uparrow\rangle$ quantum state and one with the electron in the $|\downarrow\rangle$ quantum state. The right side of the picture is similar but now the magnetic field is aligned along the x -axis (SGx). The electron that leaves the magnetic field is either in quantum state $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ or $|-\rangle = (|\uparrow\rangle - |\downarrow\rangle)/\sqrt{2}$.

- (a) Write down the quantum states $|\psi_1\rangle$, $|\psi_2\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|+\rangle$ and $|-\rangle$ as column vectors.

In a new experimental setup, the electrons in quantum state $|\downarrow\rangle$ are blocked by the black square in the figure below and only the electrons in the $|\uparrow\rangle$ quantum state reach the second magnetic field, which is aligned along the x -axis. After leaving the second magnetic field, only the electrons in quantum state $|+\rangle$ can reach the last magnetic field, aligned along z again.



- (b) What is the probability that an electron that leaves the second magnetic field (SGx) is in quantum state $|+\rangle$ or in quantum state $|-\rangle$?
- (c) What is the probability that an electron that leaves the third magnetic field (SGz-2) is in quantum state $|\uparrow\rangle$ or in quantum state $|\downarrow\rangle$?

2. Spin operators

The expectation value of the z -component of the spin moment of an $S = \frac{1}{2}$ particle can be written as

$$\langle S_z \rangle = \frac{\hbar}{2} p_{\uparrow} - \frac{\hbar}{2} p_{\downarrow}$$

in which $p_{\uparrow}$ and $p_{\downarrow}$ denote the probability for measuring the two possible outcomes of the z -component of the spin moment.

- (a) Derive the matrix representation of the $\hat{S}_z$ operator.
- (b) Calculate the probability for measuring $\hbar/2$ for the z -component of the spin moment when the quantum state of the particle is

$$|\psi\rangle = A\{ |\phi_1\rangle + |\phi_2\rangle \} \quad \text{with } \phi_1 = N_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \phi_2 = N_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

- (c) Calculate the expectation value of S_z for the particle described by quantum state $|\phi_1\rangle$

Consider a new quantum system consisting of a particle with a spin moment corresponding to $S = 1$

- (d) What is the magnitude of the spin moment?
- (e) What are the possible outcomes of measuring the z -component of the spin moment of this second particle?
- (f) Construct the matrix representation of the $\hat{S}_z$ operator for the $S = 1$ particle.

$$\hat{S}^2 |\psi\rangle = \hbar^2 S(S+1) |\psi\rangle$$

$$\hat{S}_z |\psi\rangle = \hbar M_S |\psi\rangle$$

3. Operators and eigenvalues

Consider the following matrix representations of the operators $\hat{A}$ and $\hat{B}$

$$\hat{A} = \begin{pmatrix} 3 & -i & 0 & 5 \\ i & \sqrt{2} & i & -\frac{1}{2} \\ 0 & -i & i & \sqrt{3} \\ -5 & \frac{1}{2} & -\sqrt{3} & 0 \end{pmatrix} \quad \hat{B} = \begin{pmatrix} 0 & 0 & 0 & \sqrt{3} \\ 0 & 2 & -1 & 0 \\ 0 & -1 & \frac{1}{2} & 0 \\ \sqrt{3} & 0 & 0 & 0 \end{pmatrix}$$

- (a) Which of the two operators is Hermitian and why?
- (b) Calculate the eigenvalues and eigenvectors of the Hermitian operator.
- (c) Given that a system is described by the state vector

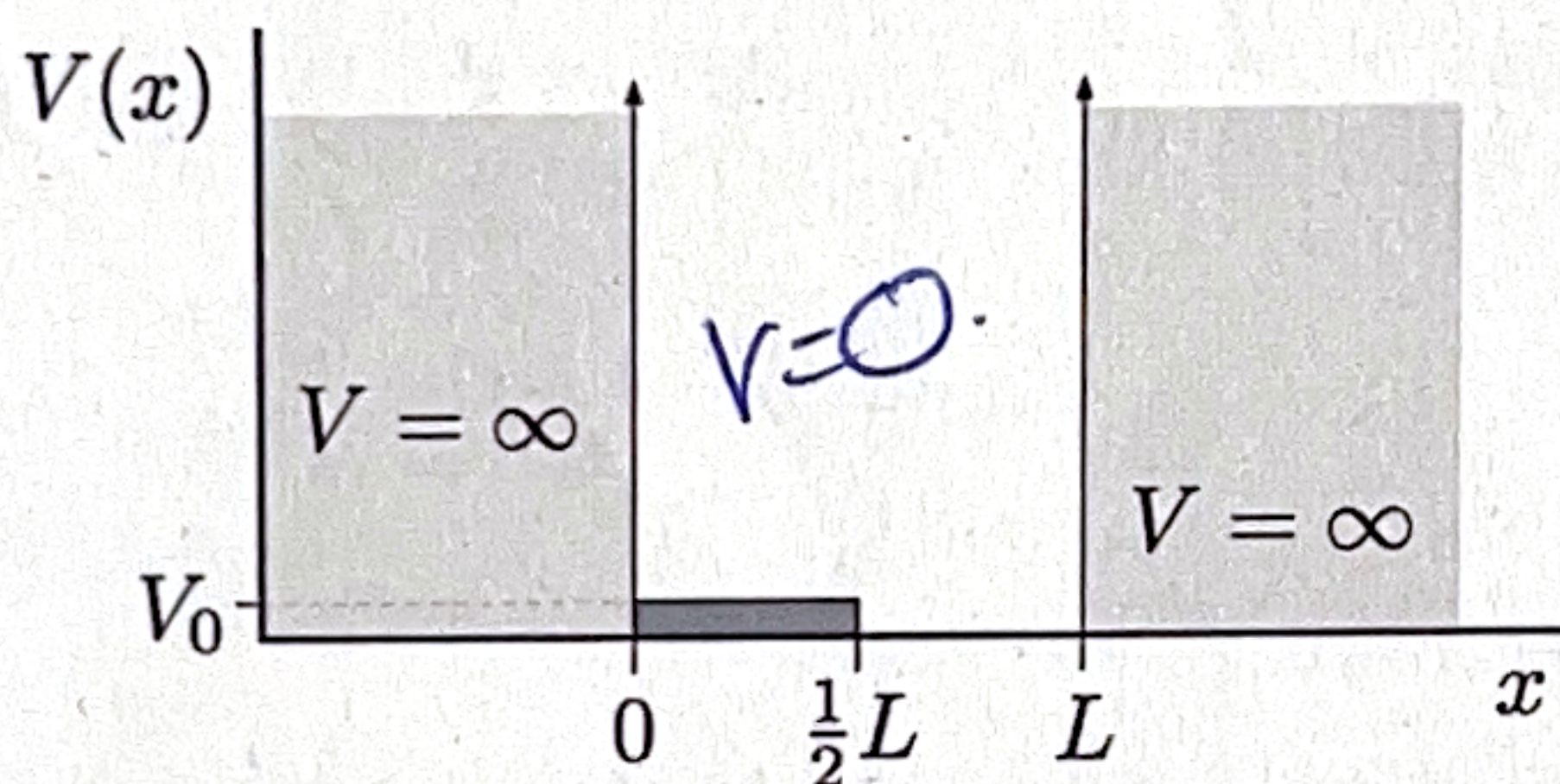
$$\psi = N \begin{pmatrix} \sqrt{3} \\ -1 \\ \sqrt{2} \\ 2 \end{pmatrix},$$

calculate the probability of measuring the highest eigenvalue of the Hermitian operator

- (d) Imagine that indeed this highest eigenvalue is measured, write down the state vector after the measurement and rationalize what outcome(s) is/are possible when a new measurement of the same observable is made immediately after the first measurement.

4. Perturbation theory S/12 S/12

- (a) Derive the expression for the energies and wave functions that describe the quantum state of a particle that is confined in a one-dimensional box of length L . The potential is zero between 0 and L , and infinite for $x > L$ and $x < 0$.
- (b) Calculate the first-order correction to the ground state energy of the particle after adding a constant potential V_0 in the interval $0 < x < \frac{1}{2}L$. Note that $V_0 \ll E$ S/6



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- The general solution of the differential equation $\frac{d^2}{dx^2} f(x) + a^2 f(x) = 0$ can be written as $f(x) = c_1 \sin(ax) + c_2 \cos(ax)$
 - $\int \sin^2 ax \, dx = \frac{x}{2} - \frac{1}{2a} \sin ax \cos ax$
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5. Formal aspects

- (a) Calculate the commutator of $\hat{P}_z$ and $\hat{Z}$
- (b) Show that the quantum state $|\Psi\rangle = N\{c_1 |\psi_1\rangle + c_2 |\psi_2\rangle\}$ is an eigenstate of operator $\hat{A}$ when $|\psi_1\rangle$ and $|\psi_2\rangle$ are eigenvectors of $\hat{A}$ sharing the same eigenvalue. 1/3
- (c) Given that (i) $[\hat{A}, \hat{B}] = 0$; (ii) the eigenvalues of both $\hat{A}$ and $\hat{B}$ are non-degenerate; and (iii) $|\phi_1\rangle$ and $|\phi_2\rangle$ are eigenvectors of $\hat{A}$, show that $|\Phi\rangle = c_1 |\phi_1\rangle + c_2 |\phi_2\rangle$ is (not) an eigenvector of $\hat{B}$.
- (d) Consider a three dimensional Hilbert space and write down the matrix representation of an anti-Hermitian operator. Illustrate with an example that the expectation values of an anti-Hermitian operator are purely imaginary.
- (e) Show that the angular momentum operators $\hat{L}_x$ and $\hat{L}_y$ do (not) commute
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$$\hat{L}_x = -i\hbar \left(\hat{Y} \frac{\partial}{\partial z} - \hat{Z} \frac{\partial}{\partial y} \right)$$

$$\hat{L}_y = -i\hbar \left(\hat{Z} \frac{\partial}{\partial x} - \hat{X} \frac{\partial}{\partial z} \right)$$

6. Harmonic oscillator

The movement of the two hydrogen atoms in an H_2 molecule can be described to a very good approximation as a quantum harmonic oscillator with a force constant of 576 N/m

- (a) Calculate the frequency ω in s^{-1}
- (b) Calculate the zero-point energy of the H_2 molecule, that is, the energy of the molecule in the limit of $T \rightarrow 0$ K.
- (c) Calculate the wave length (in nm) of a photon that promotes the H_2 molecule from its ground state to the first excited state.
- (d) Express the creation operator $\hat{a}^\dagger$ in terms of the linear impulse operator $\hat{P}_x$ and the position operator $\hat{x}$.
- (e) Using the relation $x_0 = \sqrt{\hbar/m\omega}$, write down the expression of $\hat{a}^\dagger$ in position representation
- (f) Use the equation $\hat{a}^\dagger\psi_0(x) = \psi_1(x)$ to obtain an analytical expression for the first excited state of the harmonic oscillator.

The potential energy operator: $\hat{V}(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2$

mass of an H-atom: $1.6726 \cdot 10^{-27}$ kg

reduced mass formula: $\mu = m_1m_2/(m_1 + m_2)$

$\hbar = 1.054 \cdot 10^{-34}$ kg m² / s

1 J = $6.242 \cdot 10^{18}$ eV

x eV = 1240 / x nm

$\hat{a}^\dagger = \frac{\hat{q} - i\hat{p}}{\sqrt{2}}$ $\hat{p} = \hat{P}/\sqrt{m\omega\hbar}$ $\hat{q} = \hat{x}\sqrt{m\omega/\hbar}$

$\psi_0(x) = Ae^{-x^2/2x_0^2}$